

Automated Sleep Apnea Screening

Level 3: Presentation & Business Strategy

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- **Modular Architecture:** Feature extraction, signal processing, and model inference are decoupled.
- **Robust Data Processing:**
 - Automatic sentinel value filtering (e.g., 511 for Pulse, 127 for SpO2).
 - Signal resampling to a uniform 10 Hz target frequency.
- **Performance Optimization:** Leveraging Streamlit's `@st.cache_resource` and `@st.cache_data` for fast dashboard rendering.
- **Scalability:** Easily integrates new patient EDF data into the pipeline.

Clinical Dashboard (Streamlit)

- **Automated Triage:** Instantly computes the Apnea-Hypopnea Index (AHI) and classifies severity (Normal, Mild, Moderate, Severe).
- **Interactive Signal Viewer:** Integrated Plotly charts for zooming and panning across multi-channel signals (Flow, SpO2).
- **Event Highlighting:** Clear visual markers (red overlays) for model-predicted 30-second OSA epochs.
- **User-Centric Design:** Simple dropdown navigation allows clinicians to review patient records effortlessly.

Deployment Strategy

- **Containerization:** Use Docker to bundle the Streamlit app, ML models, and system dependencies.
- **Cloud Hosting:** Deploy on Google Cloud Run or AWS Fargate for a serverless, highly-available architecture.
- **CI/CD Integration:** GitHub Actions for automated testing of data processing logic and seamless deployment upon code updates.
- **Security & Privacy:** Ensure data anonymization during ingestion and utilize encrypted storage for clinical records.

Business KPIs & Value Proposition

- **Increased Throughput:** Scales clinic capacity from 15 cases/day to 50+ cases/day.
- **Faster Turnaround:** Reduces diagnostic wait times from 3–7 days to under 10 minutes.
- **Cost Efficiency:** Projects a 60% reduction in screening costs by automating manual epoch scoring.
- **Clinical Aid:** Functions as an advanced decision-support tool, mitigating false negatives and reducing physician fatigue.

Thank You